

Machine Learning Interatomic Potential for Formic Acid Dimer (FAD)

Gabriela Bieda

288684

1. Methodology

The aim of this task was to train and apply a machine learning interatomic potential for the formic acid dimer using DeePMD kit and LAMMPS. The workflow included preparation of ORCA AIMD data, conversion into the DeePMD format, model training and testing, followed by molecular dynamics simulation with the trained potential.

The reference data originated from ab initio molecular dynamics simulations performed at two temperatures. Configurations generated at 500 K were used for training, whereas structures obtained at 298 K served as an independent validation and testing set. The molecular structure of the formic acid dimer used in this study is shown in Fig. 1. Therefore, the molecular dynamics trajectories were preprocessed and converted into the DeePMD format, generating datasets containing the information required for model training.

The model employed the `se_e2_a` descriptor to describe local atomic environments within a cutoff radius of 6 Å. The descriptor network consisted of three hidden layers with increasing numbers of neurons, while the fitting network used three larger hidden layers to predict the potential energy of the system. Both energy and force information were included in the loss function, with a stronger initial weight assigned to forces to improve the learned potential energy surface.

Training was performed for 50,000 optimization steps using an exponentially decreasing learning rate (Tab. 1). The model was trained on the 500 K trajectory and validated on the 298 K trajectory at regular intervals of 1000 steps. During training, model quality was monitored using root mean square errors calculated for both the training and validation datasets, while checkpoints were saved every 10,000 steps. After training, the final model was frozen, compressed and tested on the independent 298 K dataset containing 1001 molecular configurations. The compressed model was then used as an efficient machine learning potential in the subsequent LAMMPS simulations.

The trained potential was then applied in LAMMPS molecular dynamics simulations, which was performed in the canonical ensemble at 298 K using periodic boundary conditions (PBC) and a timestep of 0.0005 ps.

Tab. 1 DeePMD-kit model hyperparameters used in this study.

Hyperparameter	Value
Descriptor	<code>se_e2_a</code> (smooth-edition, 2-body)
Cutoff radius (<code>rcut</code>)	6.0 Å
Smooth cutoff (<code>rcut_smth</code>)	0.5 Å
Descriptor neuron layers	[10, 20, 40]
Axis neuron	4
Fitting net neurons	[100, 100, 100]
Number of training steps	50,000
Learning rate schedule	Exponential decay: 1×10^{-3} to 3.51×10^{-8}
Decay steps	5,000
Loss - energy prefactors	start: 0.02, limit: 1
Loss - force prefactors	start: 1000, limit: 1
Batch size	Auto

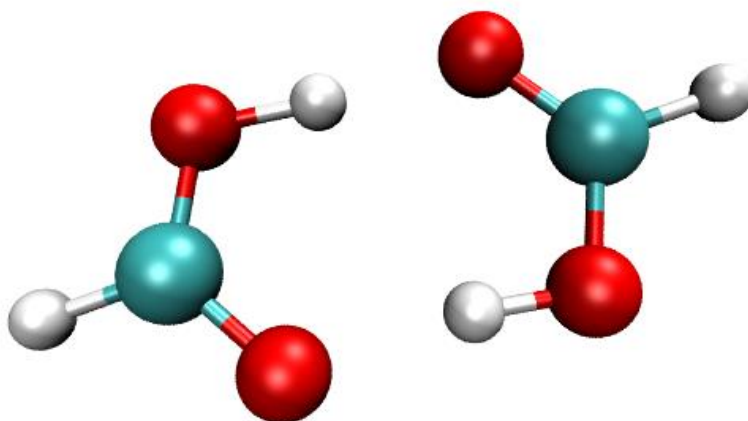


Fig. 1 Formic acid dimer used for analysis.

2. Discussion

The learning curves for both training and validation RMSE decrease overall decrease over the 50,000 training steps, indicating stable convergence without signs of overfitting. The initial RMSE of approximately 30 eV (dominated by the large force prefactor in the early loss function) drops by three orders of magnitude by the end of training (Fig 2). The close tracking of training and validation curves is particularly encouraging: the validation error (298 K) remains slightly below the training error (500 K) throughout, which suggests the 298 K structures occupy a lower energy and smoother region of the potential energy surface that the model learns more easily.

A slight plateau region is visible between steps 10,000 and 20,000, after which a further decrease occurs, likely corresponding to the transition from a force-dominated loss regime to a more balanced energy/force regime as the prefactors evolve along their exponential schedule.

Importantly, no divergence between the training and validation curves is observed at later stages of training. This indicates that the model does not memorize the training data, but retains predictive ability for configurations not used directly for optimization. The slightly lower validation RMSE can be explained by the lower temperature of the validation trajectory, where molecular geometries are less distorted and therefore easier to predict than the broader configurational space sampled at 500 K.

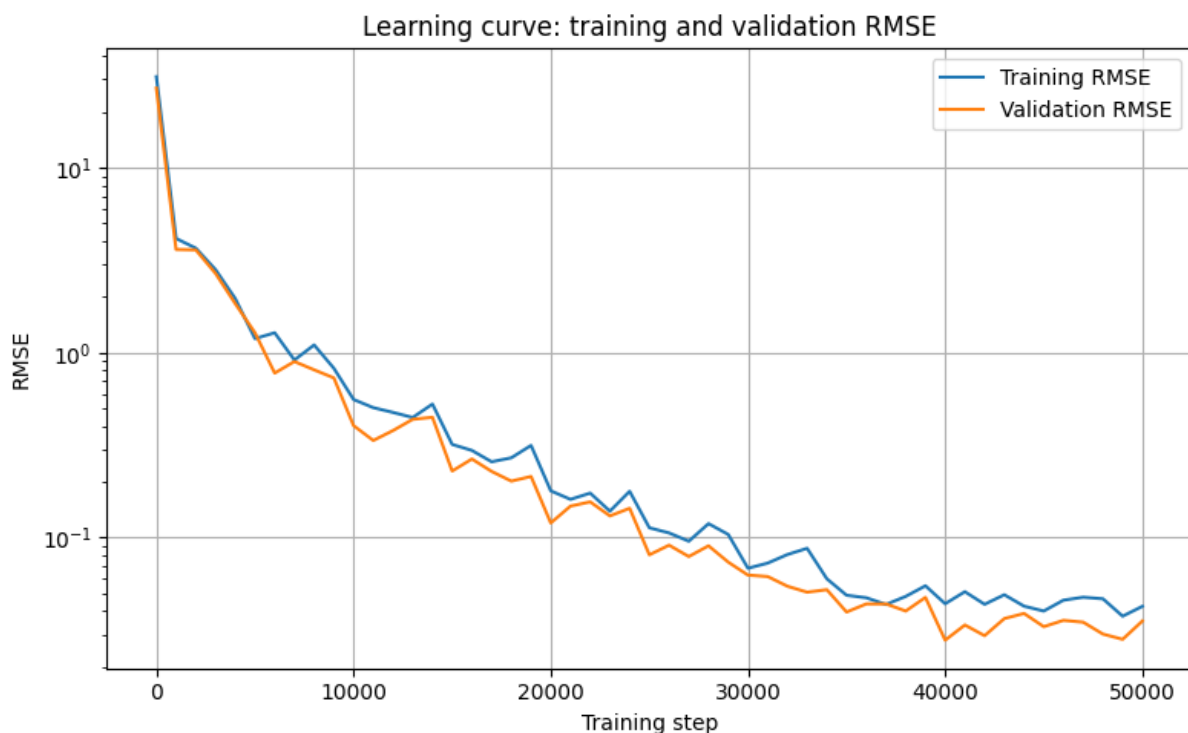


Fig. 2 Learning curve showing training and validation RMSE as a function of training step.

The force RMSE curve reveals that the model achieves a stable plateau in force accuracy after approximately 10,000 steps (Fig. 3). The training (500 K) force RMSE is consistently higher than the validation (298 K) force RMSE, which shows that at higher temperatures, atomic configurations explore more anharmonic regions of the potential energy surface with steeper and more variable forces, making them intrinsically harder to learn. The validation force RMSE of about 0.025 eV/Å indicates good agreement with the reference AIMD forces for this small molecular system.

After the initial rapid decrease, the force RMSE fluctuates around a relatively narrow range rather than decreasing continuously, suggesting that the main force patterns were learned early in the training process. Further training mainly refines the model instead of changing its behaviour substantially. This is important because accurate force prediction is essential for stable molecular dynamics simulations, as forces directly determine atomic motion during trajectory propagation.

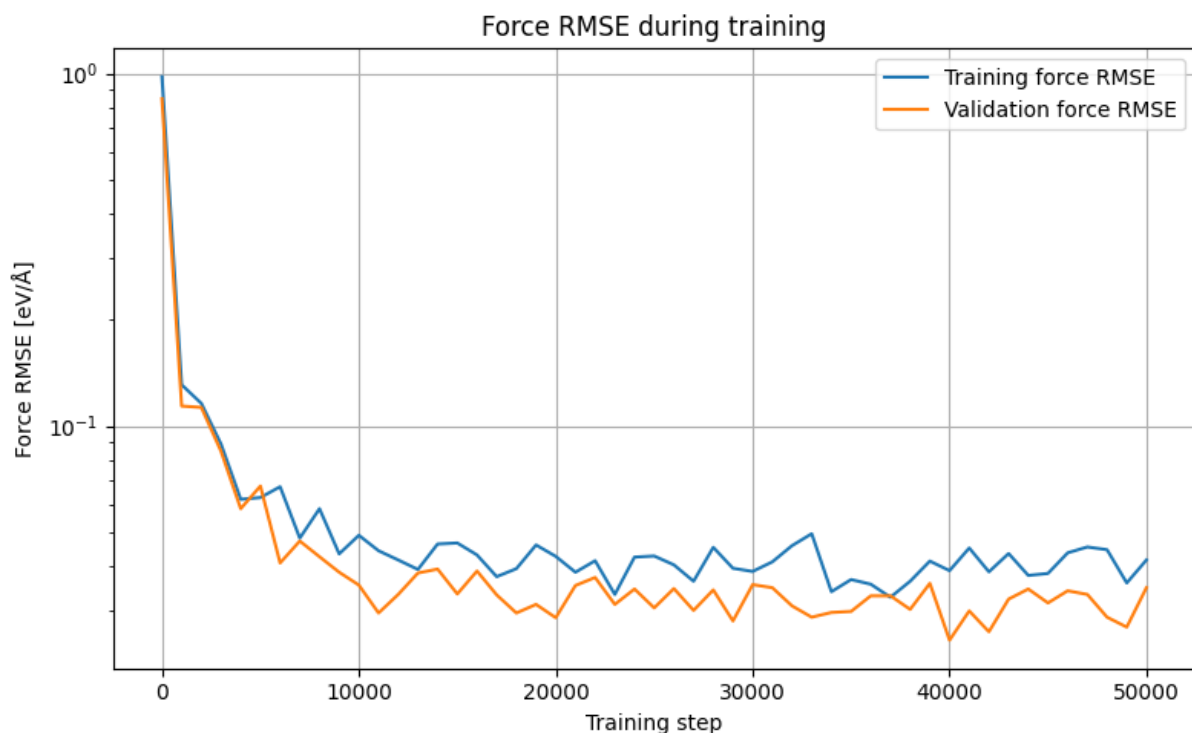


Fig. 3 Force RMSE during training for both training (500 K) and validation (298 K) datasets.

The AIMD trajectory analysis confirms that the two datasets sample different, but partially overlapping, regions of configurational space. The 500 K trajectory exhibits larger energy fluctuations, with a peak to peak amplitude of about 0.6 eV, compared with about 0.25 eV for the 298 K trajectory (Fig. 4). This indicates that the higher temperature simulation explores a broader range of molecular geometries and force environments.

The broader energy range sampled at 500 K supports its use as the training dataset, because it exposes the model to more diverse configurations. In contrast, the 298 K trajectory is more compact and energetically narrower, making it suitable as a validation set for testing whether the model can reproduce lower temperature configurations. The partial overlap between the two trajectories supports model generalization from the higher temperature training set to the lower temperature validation set.

The slight downward drift visible in both absolute energy traces over 1000 frames remains small compared with the total energy of the system and does not indicate structural instability. Overall, the energy profiles show that the selected training and validation datasets are physically related, but differ enough to provide a meaningful test of model transferability.

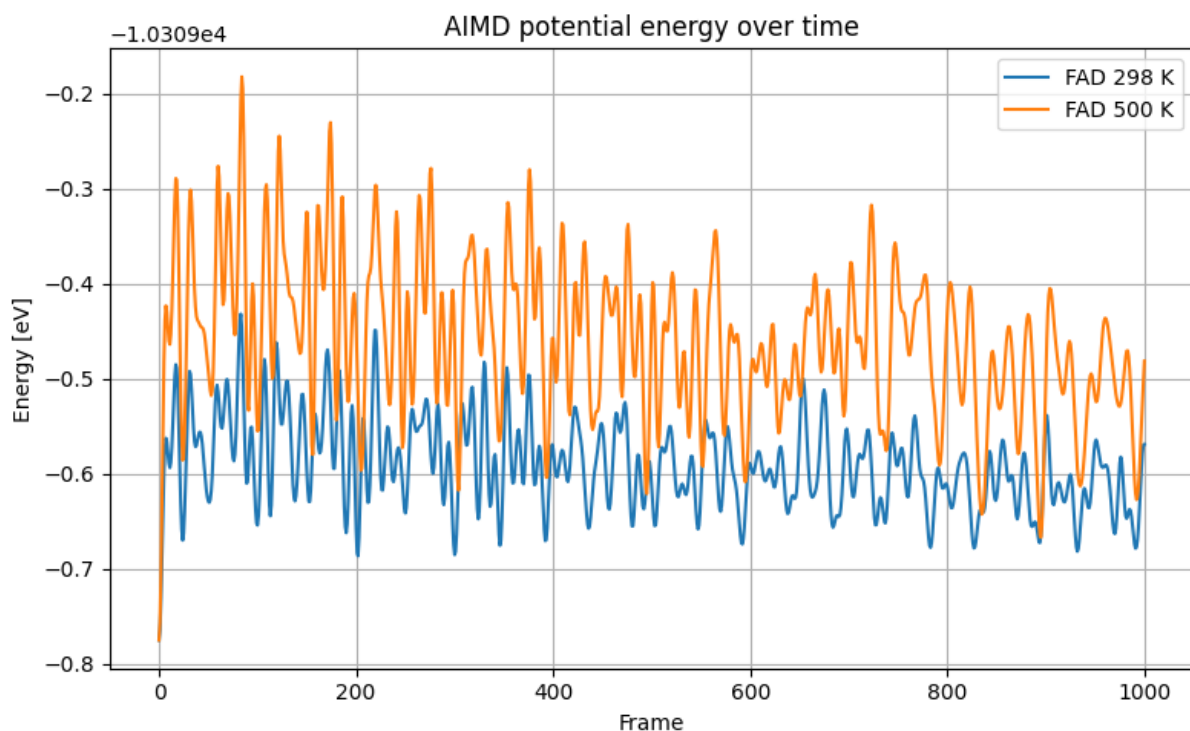


Fig. 4 Absolute AIMD potential energy as a function of simulation frame for both datasets.

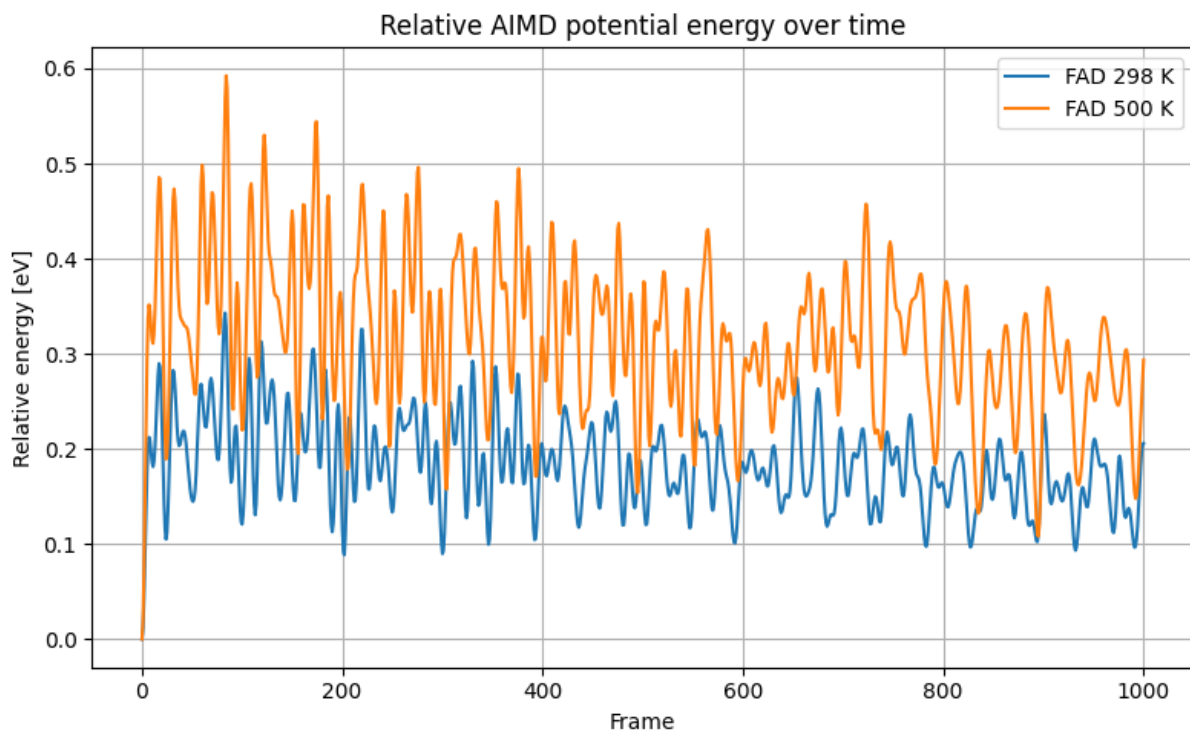


Fig. 5 Relative AIMD potential energy (shifted to zero at the minimum energy of each dataset).

Fig. 6 presents the distribution of potential energies sampled during the AIMD simulations at 298 K and 500 K. Clear differences can be observed between the two datasets. The 500 K trajectory exhibits a broader energy distribution and is shifted toward higher energies compared to the 298 K dataset. This behavior is expected because increased temperature allows the system

to access a larger portion of the potential energy surface, including higher-energy configurations that are not frequently sampled at lower temperatures.

The 298 K dataset shows a narrower and more localized distribution centered around lower energies. This indicates that the system remains closer to its equilibrium geometry and explores a more restricted configurational space. In contrast, the broader distribution at 500 K reflects increased molecular motion and larger structural deviations from the equilibrium structure.

Although the distributions are separated, a partial overlap is visible in the intermediate energy region. This overlap ensures that the validation structures are not entirely outside the configurational space sampled during training. Consequently, the trained model can be evaluated on physically related configurations while still being challenged by a dataset generated under different thermodynamic conditions.

The wider energy range observed for the 500 K trajectory further supports its selection as the training dataset. By exposing the neural network to more diverse molecular environments, the resulting model is expected to achieve better transferability and generalization when predicting the lower-temperature configurations contained in the validation dataset.

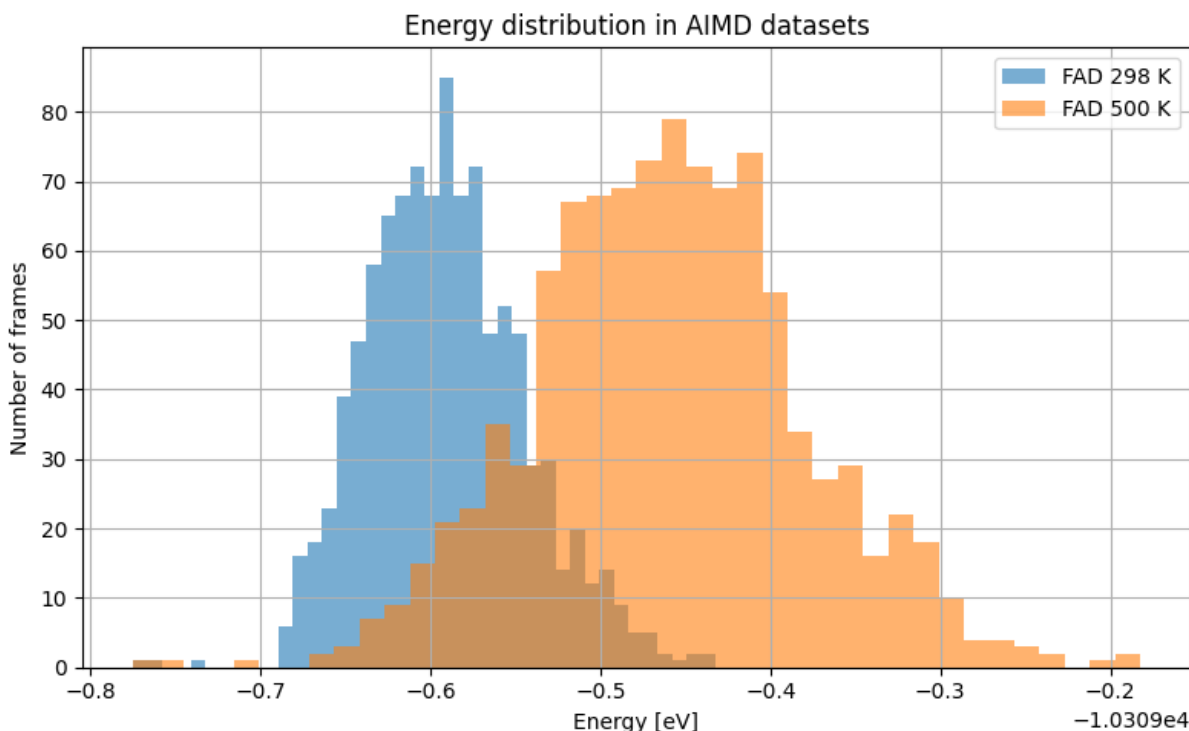


Fig. 6 Absolute potential energy distribution for the FAD AIMD datasets at 298 K (blue) and 500 K (orange).

The final model achieves an energy MAE of 2.25 meV and energy RMSE of 2.89 meV on the 298 K validation set. These values are well below 1 kcal/mol, which corresponds to approximately 43 meV and is commonly used as a reference threshold for chemical accuracy. Force errors, with MAE of 23 meV/Å and RMSE of 29 meV/Å, are also low for this molecular system and indicate that the model reproduces the reference AIMD forces with good accuracy.

The small difference between MAE and RMSE suggests that the error distribution is relatively uniform and that there are no strong outliers among the tested configurations. This is important because occasional large force errors could destabilize molecular dynamics

trajectories, even if the average error remained low. Therefore, the obtained force errors support the stability of the subsequent LAMMPS simulation.

The ratio between force and energy errors is expected, since forces are vector quantities evaluated for every atom and Cartesian direction. They provide a larger and more demanding training target than total energy alone. Overall, the quantitative errors confirm the conclusions drawn from the learning curves: the model generalizes well from the 500 K training trajectory to the independent 298 K validation set.

Tab. 2 Quantitative model performance on the FAD 298 K validation set (1001 frames), obtained using dp test on the compressed model.

Metric	Value	Unit	Notes
Energy MAE	2.25×10^{-3}	eV	FAD 298 K test set
Energy RMSE	2.89×10^{-3}	eV	
Force MAE	2.30×10^{-2}	eV/Å	
Force RMSE	2.94×10^{-2}	eV/Å	

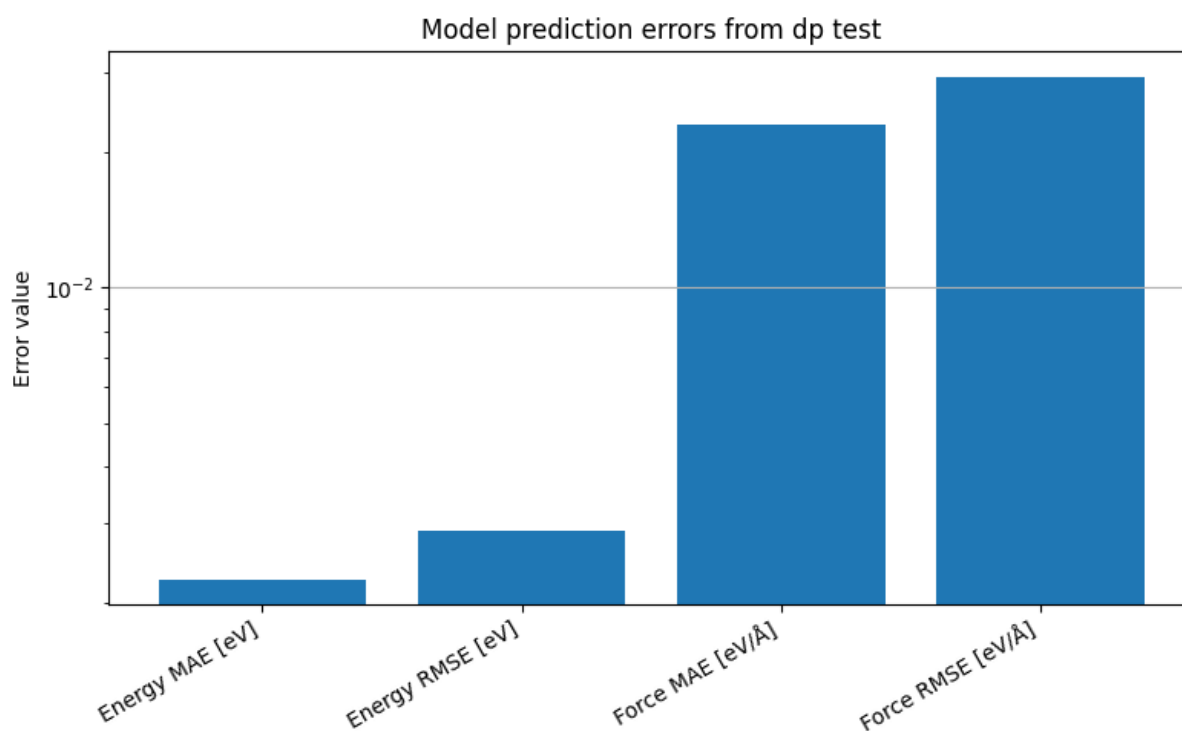


Fig. 7 Summary bar chart of model prediction errors from dp test on the FAD 298 K validation set

To further evaluate the performance of the trained machine learning potential in molecular dynamics simulations, the root mean square deviation (RMSD) relative to the initial structure was monitored throughout the LAMMPS trajectory (Fig. 8). The RMSD remains below approximately 0.26 Å during the entire 10,000-step simulation and fluctuates around an average value of about 0.15 Å. No systematic increase in RMSD is observed, indicating that the structure remains stable and does not undergo significant geometric distortions. The observed fluctuations are consistent with normal thermal motion at 298 K. These results demonstrate that the compressed DeepMD model generates a physically stable trajectory and preserves the structural integrity of the formic acid dimer during molecular dynamics simulations. The RMSD analysis is consistent with the low energy and force errors obtained from the dp test, confirming that the good predictive accuracy of the model translates into stable molecular dynamics behavior.

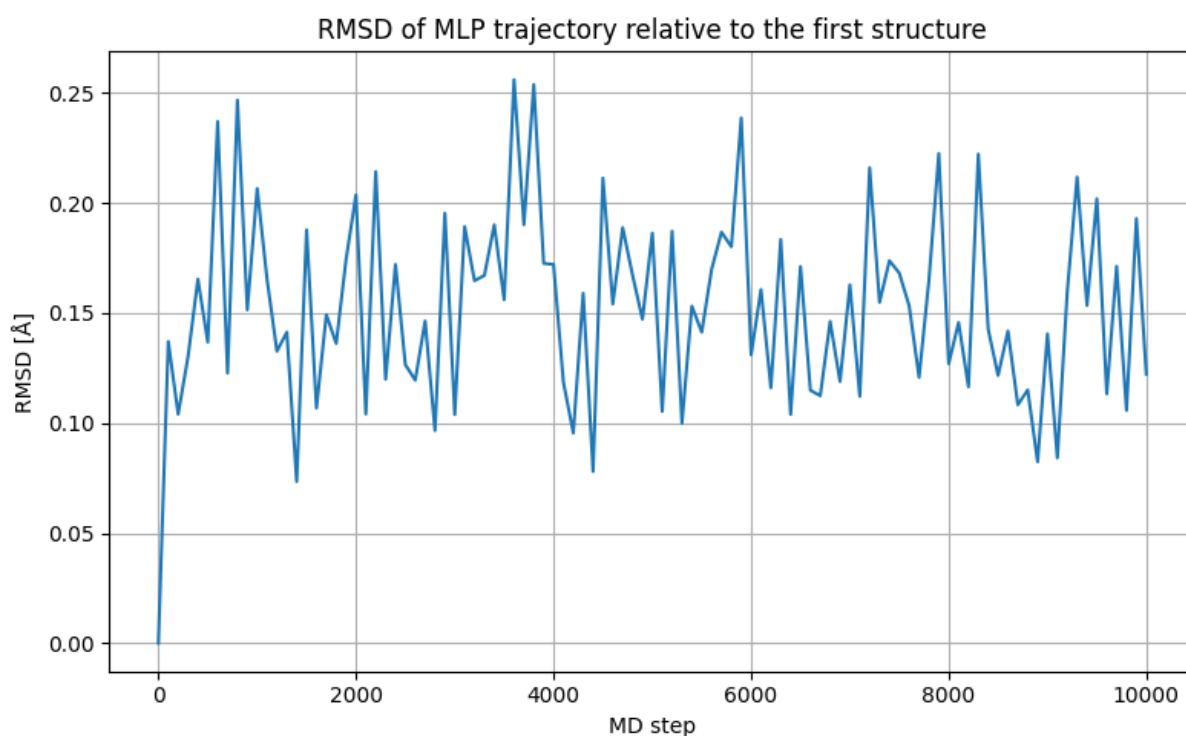


Fig. 8 RMSD of the MLP trajectory relative to the first structure during the 10,000-step LAMMPS molecular dynamics simulation at 298 K..

3. Conclusions

A machine learning interatomic potential for the formic acid dimer was successfully trained using DeepMD-kit with the `se_e2_a` descriptor and subsequently applied in LAMMPS molecular dynamics simulations. The obtained results show that the model learned the potential energy surface of the system with good accuracy and stability.

The training process showed stable convergence over 50,000 optimization steps. Training and validation RMSE values decreased throughout the optimization and remained close to each other, which indicates good generalization and no clear signs of overfitting. The final test errors were low, with an energy MAE of 2.25 meV, an energy RMSE of 2.89 meV, a force MAE of 23 meV/Å and a force RMSE of 29 meV/Å. These results confirm that the model accurately reproduces both reference AIMD energies and forces for the validation dataset.

Using the 500 K trajectory as the training set and the 298 K trajectory as the validation set was an appropriate choice. The higher temperature trajectory sampled a broader range of molecular configurations and energy values, which exposed the model to more diverse atomic environments during training. In contrast, the 298 K trajectory represented a narrower and lower energy region of the potential energy surface. The good performance on the 298 K validation set confirms that the model trained on the broader 500 K dataset was able to generalize to lower temperature configurations.

The compressed model was successfully used as an efficient machine learning potential in LAMMPS, demonstrating that the trained DeepMD model can be applied for molecular dynamics simulations of the formic acid dimer. Overall, the workflow combining ORCA AIMD data, DeepMD training and LAMMPS deployment provides a complete and reproducible approach for developing machine learning potentials for small molecular systems.

Although the model performed well, its accuracy and transferability could be further improved. The most useful improvement would be to increase the size and diversity of the training data, for example by combining 298 K and 500 K configurations into a larger training set and keeping an independent subset for validation. Additional AIMD simulations at intermediate temperatures could also improve the coverage of the potential energy surface. Increasing the number of optimization steps beyond 50,000 would probably provide only limited improvement, since the force RMSE reached a plateau relatively early. A larger neural network or descriptor architecture could be tested, but this would be most beneficial if supported by a larger and more diverse training dataset.